DEPI PROJECT DOCUMENTATION

PHASE TWO

SYSTEM ANALYSIS & DESIGN

Store Sales Data Analysis

Prepared for the Store Sales Data Analysis project

This document contains the Phase Two System Analysis & Design deliverables: problem definition, use case diagram, entity relationship diagram, data flow diagrams, dashboard wireframes, and technology stack.

1. Project Overview & Problem Definition

Project Title: Store Sales Data Analysis

Problem Statement

The project aims to analyze the Superstore Sales Dataset to understand sales performance and identify important patterns and trends. The available data contains information about orders, customers, products, sales, and locations. This data needs to be analyzed and presented clearly to help users understand business performance and make better decisions.

Project Goals

- Analyze total sales and order performance.
- Identify the best-selling products and categories.
- Compare sales across regions and customer segments.
- Analyze sales trends over time.
- Build an interactive Power BI dashboard.
- Provide useful insights to support business decisions.

Project Scope

The project focuses on data cleaning, data modeling, DAX calculations, sales analysis, insight generation, and interactive dashboard development using Power BI.

Data Coverage

The current model contains two tables: Clean Data and DateTable. Clean Data stores order-line, customer, product, sales, and location attributes. DateTable supports time-based analysis. The design reflects the current model and does not introduce separate Customer or Product tables.

2. Use Case Diagram

The Use Case Diagram describes how the primary business user interacts with the Store Sales Data Analysis system. The system is treated as an analytical Power BI solution rather than a transactional application.

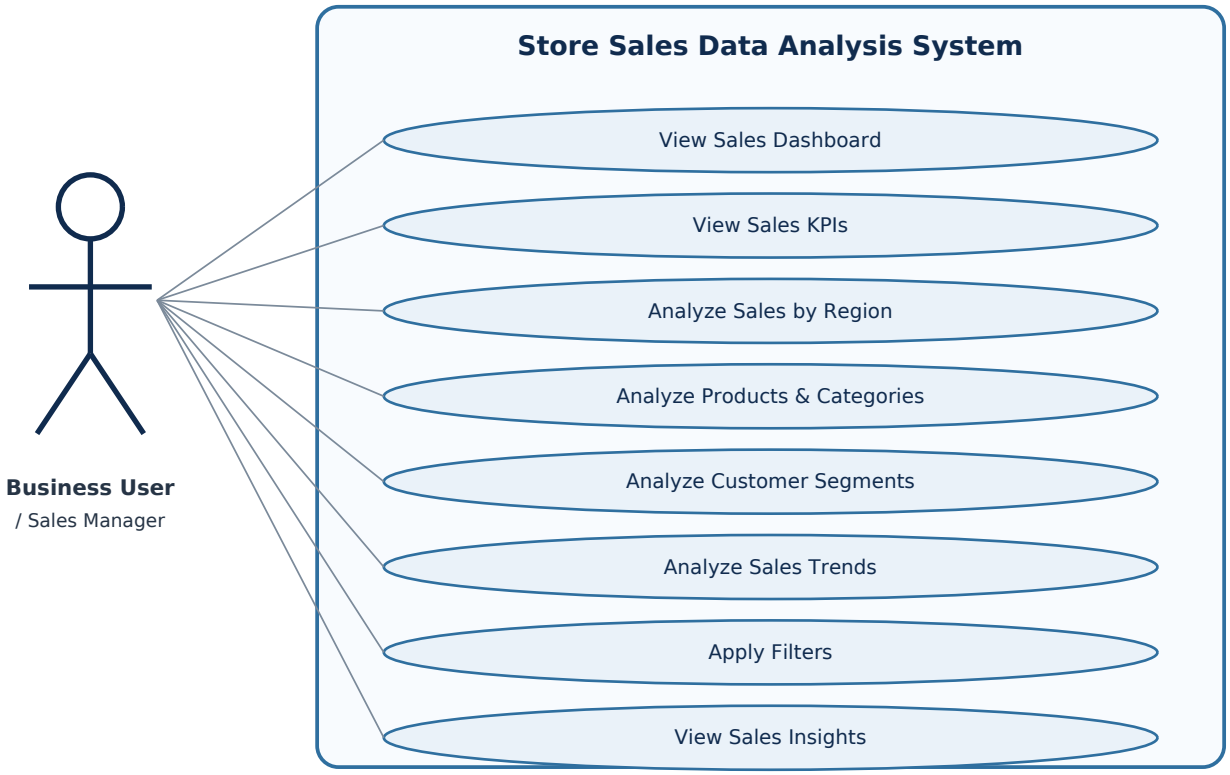


Figure 2.1 — Use Case Diagram for Store Sales Data Analysis

Primary Actor: Business User / Sales Manager

Use Cases

Use Case	Description
View Sales Dashboard	Open the sales report and navigate its pages.
View Sales KPIs	Review key sales and order indicators.
Analyze Sales by Region	Compare sales performance across regions.
Analyze Products & Categories	Explore product and category sales performance.
Analyze Customer Segments	Compare sales across customer segments.
Analyze Sales Trends	Review sales performance over time.
Apply Filters	Filter the report by available dimensions.
View Sales Insights	Review findings derived from the analysis.

Final report features should be checked against the implemented dashboard during testing.

3. Entity Relationship Diagram (ERD)

The ERD documents the current Power BI data model using two entities. It shows the key fields and the one-to-many relationship used for time-based analysis.

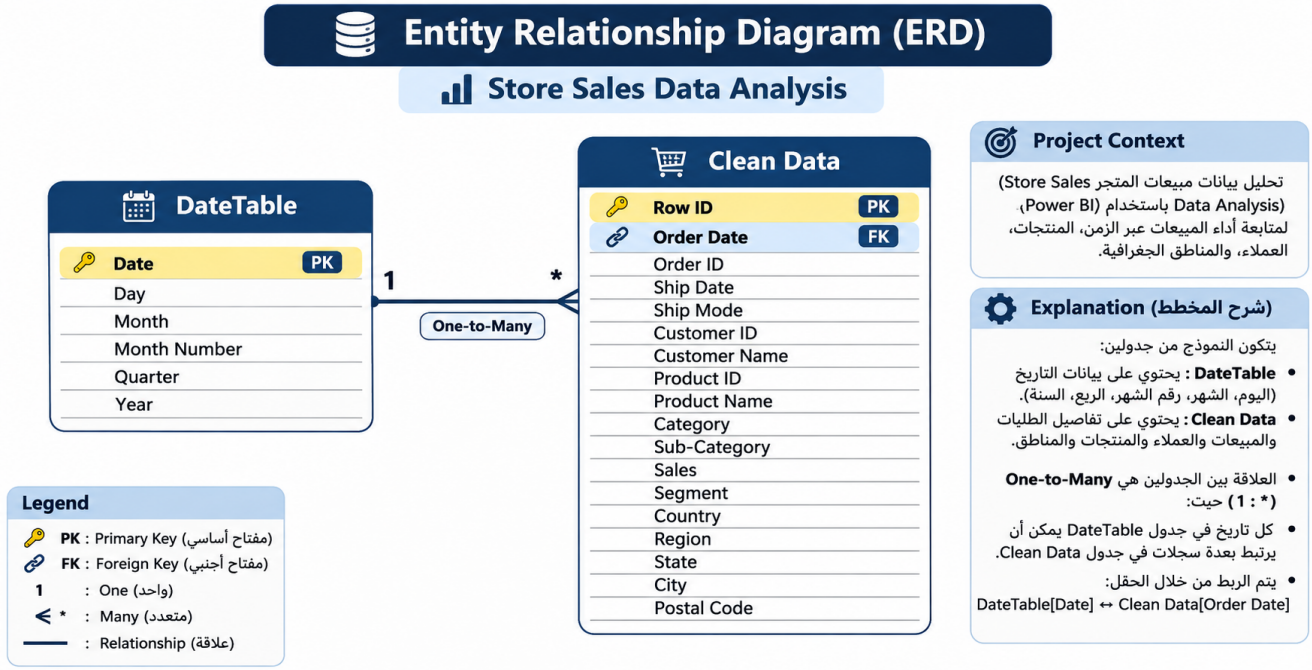


Figure 3.1 — ERD concept for the current Store Sales Data Analysis model

DateTable

Attributes: Date (PK), Day, Month, Month Number, Quarter, Year.

Clean Data

Attributes: Row ID (PK), Order ID, Order Date (FK), Ship Date, Ship Mode, Customer ID, Customer Name, Product ID, Product Name, Category, Sub-Category, Sales, Segment, Country, Region, State, City, Postal Code.

Relationship

DateTable[Date] (1) → Clean Data[Order Date] (*). One date in DateTable can be associated with multiple sales records in Clean Data. Row ID is treated as the row-level primary key based on the full-column profiling result previously reviewed.

Modeling note: Order ID, Customer ID, and Product ID are repeated in the transaction-level table and are not primary keys in this current model. Separate Customer and Product entities are intentionally not added.

4. Data Flow Diagram (DFD)

The DFD explains how source sales data is brought into the analysis workflow, prepared, modeled, analyzed, and presented to the business user.

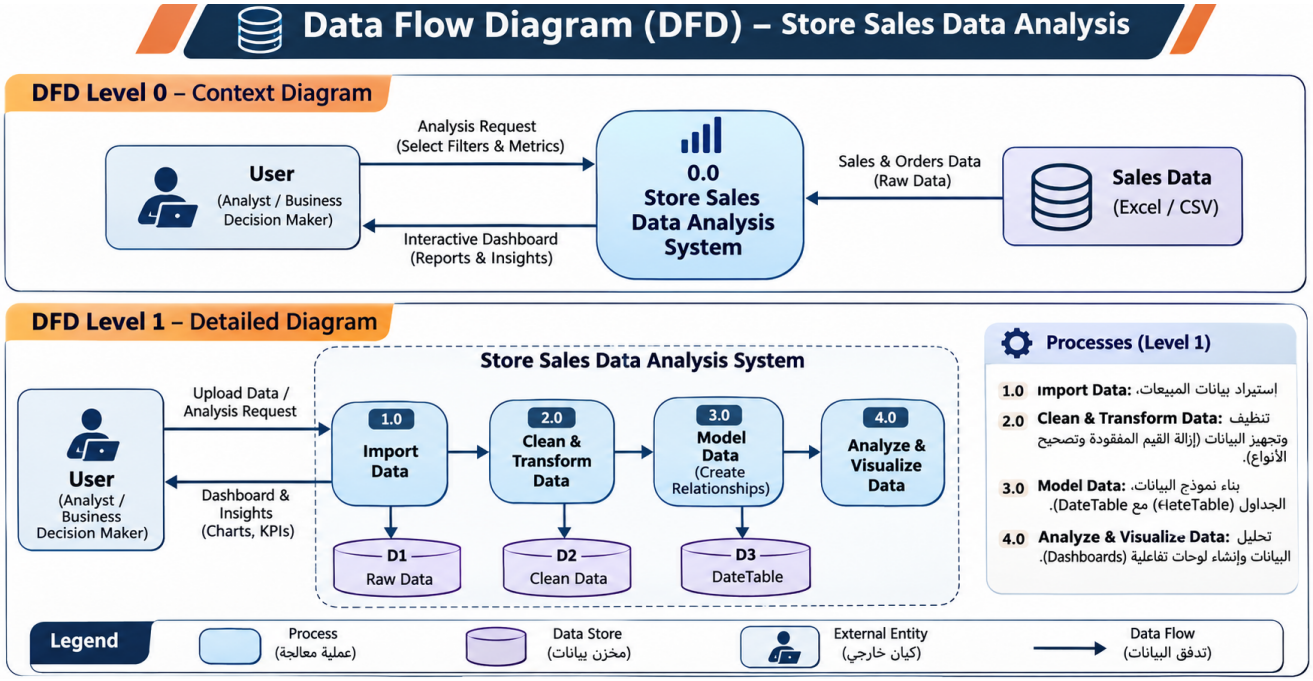


Figure 4.1 — DFD Level 0 and Level 1 overview

DFD Level 0 — Context Diagram

The system receives sales data from a structured source such as Excel or CSV. A business user interacts with the analysis system through analysis requests and filters. The system returns dashboard views, reports, and insights.

DFD Level 1 — Main Processes

Process	Responsibility
1.0 Import Data	Load the available sales dataset.
2.0 Clean & Transform Data	Inspect and prepare fields, values, and data types.
3.0 Model Data	Organize the model and create the DateTable relationship.
4.0 Analyze & Visualize Data	Create supported KPIs, charts, and interactive report pages.

D1/D2/D3 in a conceptual workflow represent raw data, cleaned data, and the date dimension; they are not three separate physical tables in the current PBIX model.

5. Dashboard Wireframes

The wireframes define the initial structure and placement of report elements. They are layout references, not the final visual design.

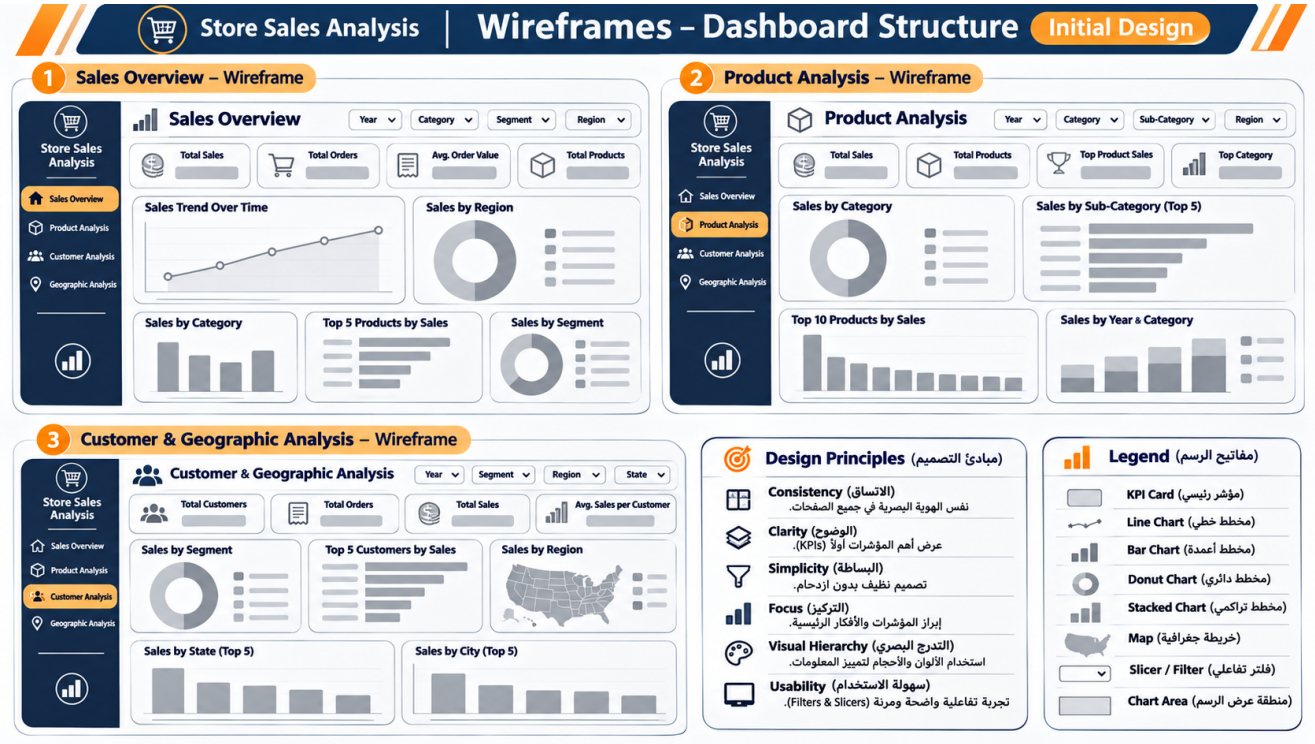


Figure 5.1 — Initial wireframe structure for the three dashboard pages

Page 1 — Sales Overview

Filters at the top; KPI cards beneath the page title; sales trend and regional analysis in the middle; category, top-product, and segment visuals in the lower area.

Page 2 — Product Analysis

Product-focused KPIs, sales by category, top sub-categories, top products, and sales by year and category.

Page 3 — Customer & Geographic Analysis

Customer and order KPIs, sales by segment, top customers, sales by region, and state/city comparisons.

Design Principles

Consistency, clarity, simplicity, focus, visual hierarchy, and usability. Each visual must be supported by available data and validated before final delivery.

6. Technology Stack

The project uses the following tools to support data preparation, modeling, analysis, reporting, and collaboration.

Technology	Role in the Project
Microsoft Excel	Source data storage and initial inspection.
Power Query	Data cleaning, transformation, and preparation.
Power BI Desktop	Data modeling, analysis, and interactive dashboard development.
DAX	Creation of analytical measures and KPIs.
GitHub	Project file sharing, version control, and collaboration.

End-to-End Workflow

Excel / CSV source → Power Query preparation → Power BI data model → DAX measures → Interactive dashboard → Insights and final reporting.

Implementation Reminder

Only tools actually used in the delivered project should be listed as implemented technologies. Any proposed or unused tool should be identified as planned rather than completed.

PHASE TWO SUMMARY

System Analysis & Design establishes the project problem, user interactions, data structure, data flow, dashboard layout, and technology choices. These design outputs provide the blueprint for the Implementation phase.